

SHAIKH MUHAMMAD ZAIN

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SUMMARY

AI-specialized Computer Science student with applied experience in machine learning, retrieval-augmented generation, and end-to-end ML system deployment, spanning supervised learning pipelines, API-based model serving, and CI/CD-driven delivery.

SKILLS

- AI & Machine Learning:** Machine Learning, Deep Learning, Retrieval-Augmented Generation (RAG), Supervised Learning (Regression, Classification, Clustering), Model Evaluation
- Core Tools:** Python, PyTorch, TensorFlow, Scikit-learn, FAISS, FastAPI, Flask, Streamlit, Pandas, NumPy
- Core CS:** Algorithms & Data Structures, Distributed Systems, Computer Networking
- Soft Skills:** Team Collaboration, Problem Solving, Communication

EXPERIENCE

Machine Learning Intern

Jul 2026 – Aug 2026

FlyRank

Remote

- Implemented supervised learning workflows on real ML project tasks over a two-month internship, covering data preparation through model evaluation.
- Earned FlyRank's Machine Learning and AI Fluency certifications, validating applied proficiency across the ML development lifecycle.

Information Technology Intern

Feb 2026 – Mar 2026

Karachi Development Authority (KDA)

Karachi, Pakistan

- Resolved day-to-day hardware and software issues for the IT department, coordinating with the wider team in a government office environment.
- Investigated real-world IT infrastructure operations at organizational scale, complementing academic coursework with practical systems experience.

PROJECTS

Semantic Search Engine over Git History (RAG + FAISS)

Python | GitHub

- Implemented a hybrid retrieval pipeline combining BM25, LSA, and sentence-transformer embeddings with FAISS for dense vector search over commit history.
- Deployed citation-checked answer generation so retrieved results are traceable back to their source commits, reducing unverifiable responses; evaluated on the huggingface/datasets repository.

EduPulse: Student Performance Intelligence Platform

Python, FastAPI, Streamlit | GitHub

- Developed an early-warning risk-scoring and score-prediction system with SHAP-based explainability and a fairness audit layer to surface biased predictions.
- Deployed the platform end-to-end via a FastAPI backend and Streamlit interface, automated with a CI/CD pipeline for testing and deployment.

Final Year Project: Hallucination-Detection Middleware for Agentic RAG Systems

Python, PyTorch, Hugging Face Transformers,

LangGraph

- Co-designed a middleware architecture intended to wrap LangGraph/LangChain agents and flag step-level hallucinations without modifying the underlying agent, using LLM log-probability screening plus a fine-tuned ModernBERT classifier.
- Proposed an automatic repair loop, a capability existing hallucination detectors lack; success criteria (F1 of 0.80+, fault localization above 72%) are proposed targets to be validated as implementation begins. Proposal approved by the FYP panel; not yet reflected in a public repository.

Rule-Based AI Chatbot

Python, Flask | GitHub

- Implemented a deterministic, rule-based conversational chatbot in Python with a Flask web interface for live interaction and testing.

EDUCATION

NED University of Engineering and Technology

Oct 2023 – Aug 2027

BS Computer Science and Information Technology (Specialization: Artificial Intelligence)

Karachi, Pakistan

DJ Sindh Government Science College

2022

Intermediate (Pre-Engineering)

Karachi, Pakistan

CERTIFICATIONS

- Google AI Professional Certificate | Coursera | Jun 2026
- FlyRank Machine Learning Certification | FlyRank | 2026
- FlyRank AI Fluency Certification | FlyRank | 2026
- Web Development Course | Aptech Computer Education